HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science

Fall 2024 Midterm Examination

Date: 18 October 2024 Time: 19:00–21:00

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed book exam. It consists of **9** questions. You are allowed to bring an A4 single-sided or double-sided, handwritten or printed, cheat sheet.
- Please write your name, student ID, email, lecture section in the space provided at the top of this page.
- Please sign the honor code statement on page 2.
- Answer all the questions within the space provided on the examination paper. You may also use the back of the pages. Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines.
- Solutions can be written in terms of binomial coefficients, factorials, and the $C(n,k)$, $P(n,k)$ notation. For example, you can write $\binom{5}{3}+\binom{4}{2}$ instead of 16. Avoid using nonstandard notations such as ${}_nP_k$ or ${}_nC_k$. Calculators are not allowed for the exam.

Problems	1	2	3	4	5	6	7	8	9	Total
Points	15	12	10	12	12	14	5	8	12	100
Score										

As part of HKUST's introduction of an honor code, the HKUST Senate has recommended that all students be asked to sign a brief declaration printed on examination answer books that their answers are their own work, and that they are aware of the regulations relating to academic integrity. Following this, please read and sign the declaration below.

I declare that the answers submitted for this examination are my own work.

I understand that sanctions will be imposed, if I am found to have violated the University regulations governing academic integrity.

Student's Name: _____

Student's Signature: _____

Problem 1: Propositional logic, Inference and Proofs [15 points]

(a) **[2 points]** Briefly argue whether

$$(p \to q) \vee (q \to r)$$

is a tautology or not. If Yes, prove it. If No, provide a counter example. (You are **not** allowed to use truth table)

(b) **[4 points]** Consider these two statements:

$$p \to (q \to r)$$
$$(p \to q) \to r$$

If they are equivalent prove it. Otherwise, provide a counter example. (You are **not** allowed to use truth table)

(c) **[6 points]** Let p, q, r, s, t denote five propositions. You are given the following set of premises:

1. $s \vee \neg t$
2. $p \to (q \wedge \neg r)$
3. $s \to p$
4. $q \to r$

Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion $(\neg p \wedge \neg t)$ is true. **Hint:** Start by combining premises 4 and 2.

(d) **[3 points]** Assume there are 13 people sitting around a round table. Construct a proof by contradiction to show that there can not be more than 6 persons between any two people.

Answer: (a)

$$\begin{aligned} & (p \to q) \vee (q \to r) \\ \equiv\ & (\neg p \vee q) \vee (\neg q \vee r) \\ \equiv\ & \neg p \vee q \vee \neg q \vee r \\ \equiv\ & \neg p \vee (q \vee \neg q) \vee r \\ \equiv\ & \neg p \vee T \vee r \\ \equiv\ & \text{True} \end{aligned}$$

(b) These two are not equivalent. Consider the following counter example:

$$p = F, q = T, r = F$$

The top equation is True however the bottom equation is False.

(c)

1. $s \vee \neg t$
2. $p \to (q \wedge \neg r)$
3. $s \to p$
4. $q \to r$
5. $\neg q \vee r$ from 4
6. $\neg(q \wedge \neg r)$ from 5
7. $\neg(q \wedge \neg r) \to \neg p$ from 2
8. $\neg p$ from 7,2
9. $\neg p \to \neg s$ from 3
10. $\neg s$ from 9,8
11. $\neg t$ from 10,1
12. $\neg p \wedge \neg t$ from 11,8

(d) Assume that there exist two people such that their distance is more than 6, i.e., there are at least 6 people sitting between them on both sides. This implies that we have at least $2 \cdot 6 + 2 = 14$ people, which is a contradiction as we have 13 people.

Problem 2: Predicate logic [12 points]

(a) **[3 points]** Consider the following predicates:

$$Prime(x) : x \text{ is prime}$$

$$Even(x) : x \text{ is even}$$

The domain is all natural numbers $(0, 1, 2, \cdots)$. Express the following statement in English:

$$\forall x((Even(x) \wedge x > 2) \rightarrow \exists y \exists z(Prime(y) \wedge Prime(z) \wedge (y + z = x)))$$

(b) **[4 points]** Negate the statement in (a) so that all negation symbols immediately precede predicates.

(c) **[5 points]** Either prove that the two statements below are logically equivalent or give a counterexample to show that they are not logically equivalent.

$$\exists y[P(y) \rightarrow \forall x Q(x)]$$

$$\exists y P(y) \rightarrow \forall x Q(x)$$

Answer: (a) ***"Every even natural number greater than 2 is the sum of two prime numbers."***

In reality this is a famous open problem called Goldbach's conjecture! We still have no proof for it and we don't know if its really correct =)

(b) The statement in part (a) can be also written like this:

$$\forall x(\neg(Even(x) \wedge x > 2) \vee \exists y \exists z(Prime(y) \wedge Prime(z) \wedge y + z = x))$$

Here is the negation of this statement:

$$\exists x((Even(x) \wedge x > 2) \wedge \forall y \forall z \neg(Prime(y) \wedge Prime(z) \wedge y + z = x))$$

Which also can be written as:

$$\exists x((Even(x) \wedge x > 2) \wedge \forall y \forall z(\neg Prime(y) \vee \neg Prime(z) \vee y + z \neq x))$$

(c) These two statements are obviously not equivalent. Consider this scenario, Imagine having $\{a, b\}$ in our domain. Assume $P(a) = T, P(b) = F, Q(a) = T, Q(b) = F$. The top statement is True. Because $P(b) = F$ which makes the top statement True. However the bottom statement becomes False.

Problem 3: Sets [10 points]

(a) **[4 points]** Determine whether each of the following infinite sets is countable or uncountable. No explanation is needed.

(i) **[1 point]** The set of all java programs!

(ii) **[1 point]** The set of coordinates (X, Y) in 2D space such that $X \in Q$ (X is rational number) and $Y \in Q'$ (Y is irrational)

(iii) **[1 point]** The set of coordinates (X, Y) in 2D space such that both X, Y are rational numbers.

(iv) **[1 point]** The set $\mathcal{P}(\mathbb{R}) - \mathbb{R}$, where $\mathbb{R}$ is the set of real numbers and $\mathcal{P}(\mathbb{R})$ is its power set.

(b) **[2 points]**

(i) **[1 point]** Let $\mathbb{R}$ be the set of real numbers. Provide an example for an infinite set B such that $\mathbb{R} - B$ is **countable infinite**.

(ii) **[1 point]** Let $\mathbb{R}$ be the set of real numbers. Provide an example for an infinite set B such that $\mathbb{R} - B$ is **uncountable**.

(c) **[4 points]** Prove that $A - (B \cap C) = (A - B) \cup (A - C)$. You are **Not** allowed to use Venn diagram to prove it.

Answer: (a) (i) Countable

(ii) Uncountable

(iii) Countable

(iii) Uncountable

(b) (i) $B = \mathbb{R} - \mathbb{N}$

(ii) $B = (0, 1)$

(c) Assume $x \in A - (B \cap C) \leftrightarrow x \in A \wedge x \notin (B \cap C) \leftrightarrow x \in A \wedge (x \notin B \vee x \notin C) \leftrightarrow (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) \leftrightarrow (x \in (A - B)) \vee (x \in (A - C)) \leftrightarrow x \in ((A - B) \cup (A - C))$

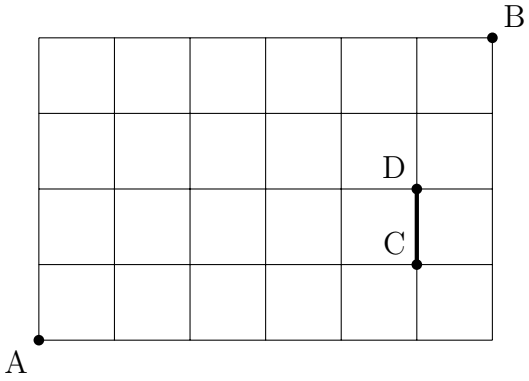
Problem 4: Counting I [12 points]

(a) **[6 points]** In the following picture, starting from A we wish to reach B, by taking edges *Up* or *Right*. How many different paths (two paths are different if they differ in at least one edge) exist from A to B such that:

(i) **[2 points]** There are no restrictions. No explanation is needed. **Hint:** Any path from A to B is a combination of *Up* and *Right* moves for a total of 10.

(ii) **[2 points]** You have to pass from CD edge. No explanation is needed.

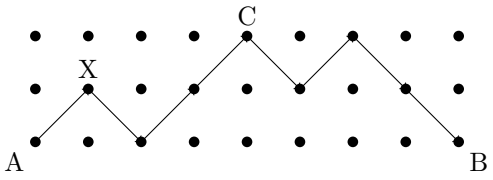
(iii) **[2 points]** You can not pass from CD edge. No explanation is needed.



(b) **[6 points]** Consider the following picture. You are only allowed to move ↗ (one to the right and up) and ↘ (one to the right and down) only. In how many ways you can get from A to B such that:

(i) **[3 points]** There are no restrictions. Briefly explain why. Simply counting all possible ways will not give you full credits.
Hint: Note that in order to get from A to B you have to pass from some certain points like X.

(ii) **[3 points]** You want to pass from C. Briefly explain why. Simply counting all possible ways will not give you full credits.



As you can see in this picture, This is a valid path from A to B.

Answer: (a) (i) $\binom{10}{4}$ Alternative explanation: We have to select with repetition 4 Up edges from the 7 columns: $\binom{7+4-1}{4} = \binom{10}{4}$

(ii) $\binom{6}{1} \cdot \binom{3}{1}$

(iii) $\binom{10}{4} - \binom{6}{1} \cdot \binom{3}{1}$

(b) (i) 2^3 . Note that the path has to pass from the middle point on even columns. So for columns with odd index (from 3) we have two options to either go up or down. Thus the answer is 2^3

(ii) We can simply see that for the fifth column we have to move up. So the total possible ways is 2^2.

Problem 5: Counting II [12 points]

(a) **[6 points]** How many ways are there to place numbers 1 to 6 on a dice so that different rotations are considered equivalent? Explain your solution.

(b) **[6 points]** In how many ways we can pick 3 different numbers from $\{1, 2, 3, \cdots, 2n\}$ such that they form arithmetic sequence? Explain your solution. (Arithmetic sequence is a sequence of numbers such that the difference from any succeeding term to its preceding term remains constant throughout the sequence, For example $1, 4, 7$ is an arithmetic sequence but $1, 5, 10$ is not)

Hint: Consider your sorted sequence to be a, b, c. What can you say about the oddity of a and c?

Answer: (a) **Solution 1:** Lets say we put "1" on top side. for the opposite side we have 5 possible ways. Now for the 4 remaining values on its side we have $(4-1)!$ different ways to put the numbers as they are in a round situation. So the total answer is $5 \cdot 3!$.

Solution 2: we have 6! permutations in general but each case can be duplicate if it can be made from different rotations. So we can fix one side to be on top which as 6 options. And we can now rotate it 4 times clockwise. So we have to divide 6! by $6 \cdot 4$

(b) As $b = \frac{a+c}{2}$, it means $a + c$ must be even. So they are either both odd or both even. b can be uniquely calculated when we have both a and c. Picking two different odd numbers can be done in $\binom{n}{2}$ ways and picking 2 different even numbers also can be done in $\binom{n}{2}$ ways. So the total possible ways are $\binom{n}{2} + \binom{n}{2} = 2\binom{n}{2}$

Problem 6: Counting III [14 points]

Assume that in any decimal number the leftmost digit **cannot** be 0. How many 6-digit numbers exist such that their digits from left to right:

(a) [**4 points**] Are strictly decreasing (e.g., 864320). Briefly explain why.
Hint: Any six different digits give you a unique number with strictly decreasing digits.

(b) [**3 points**] Are strictly increasing (e.g., 136789). Briefly explain why.

(c) [**4 points**] Are deceasing, but potentially contain repeated digits (e.g., 997770). Briefly explain why.
Hint: Only the number of occurrences of each digit is important.

(d) [**3 points**] Are increasing, but potentially contain repeated digits (e.g., 111144). Briefly explain why.

Answer: (a) By choosing any 6 digits from $\{0, 1, 2, \cdots, 9\}$ we can form a unique 6 digit number that its digits is strictly decreasing from left to right. so the answer is $\binom{10}{6}$

(b) Similar to part (a), but we cant pick 0 because then it has to be the leftmost digit which is impossible. So the answer is $\binom{9}{6}$

(c) We can convert this question to another question so that we count the frequency of each digit. In other words: $x_0+x_1+x_2+\cdots+x_9 = 6$. this equation has $\binom{15}{9}$ non-negative integer solutions. However we need to remove 000000 case which gives us $\binom{15}{9} - 1$

(d) Similar to part (c). However we should not pick any 0. This means $x_1 + x_2 + \cdots + x_9 = 6$ which has $\binom{14}{8}$ different non-negative integer solutions.

Problem 7: Inclusion-Exclusion [5 points]

Assume that there are 5 processing jobs that must be assigned to 3 servers.

Use Inclusion-Exclusion to calculate how many assignments exist such that each server is assigned to at least one job? Briefly explain why.

Answer: **Solution 1:** This is the number of onto functions from a set of 5 elements to a set of 3 elements: $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5$

Solution 2:

Problem 8: Pigeonhole Principle [8 points] Assume 16 teams. Each team plays exactly 5 games. A win gains 3 points, draw gains 1 point and loss gains 0 point. Prove that at least two teams will have the same number of points.

Answer: If a team with 15 points exists, the next team must have at least a loss, which means the next possible value for total points is 12 (i.e., 13 and 14 cannot occur). If no team achieves 15, the next maximum is 13 points for team(s) with for 4 wins and a tie (i.e., 14 and 15 cannot occur). In any case, the options for total points is less than 15, while we have 16 teams.

Problem 9: Subsets and Combinatorial Proof [12 points]

(a) **[7 points]** Consider the set $\{1, 2, \cdots, 2n-1\}$. Show that the number of subsets with at most $n-1$ elements is equal to $\frac{2^{2n-1}}{2}$.

Hint: Think about the size of any random subset and the size of its complement.

(b) **[5 points]** Use combinatorial argument to prove:

$$2^{2n-2} = \sum_{r=0}^{n-1} \binom{2n-1}{r}$$

Hint: You may use part (a), even if you do not manage to prove it.

Answer: For the first part note that the total number of subsets of $\{1, 2, \cdots, 2n-1\}$ is 2^{2n-1}. For each subset with at most $n-1$ elements, there is a complementary subset with n or more elements. Since the total number of subsets is 2^{2n-1} and they come in complementary pairs, exactly half, $\frac{2^{2n-1}}{2}$, have at most $n-1$ elements.

For the second part, the LHS is the number of subsets of $\{1, 2, 3, \cdots, 2n-1\}$ with at most $n-1$ elements. The RHS counts the number of subsets that have exactly 0 elements, 1 element,2 elements , $\cdots$, $n-1$ elements separately and adds them together.

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